

Forward-Looking Earnings Dominate Cross-Firm Graphs in Daily S&P 500 Volatility Forecasting

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Abstract. A recent literature wires stocks together with graph neural networks and reports that cross-firm structure improves volatility forecasts. We test the claim on daily Parkinson variance for the S&P 500 under a leakage-controlled protocol, scored on QLIKE with date-clustered Diebold–Mariano tests, multiple seeds, and full walk-forward folds. Conditional on own-history and a market factor, a correlation graph adds no reliable incremental value at any horizon: it ties a uniform-adjacency market aggregate ($h1\ p = 0.84$) and a random-edge placebo ($h1\ p = 0.13$), and it hurts at the monthly horizon. A short-horizon graph signal appears only on a minimal three-lag HAR base and vanishes once own-history momentum features enter the model, so the graph is redundant with the stock’s own past. The lever that works is forward-looking and per-firm: a scheduled-earnings feature added to a QLIKE-loss gradient-boosting model cuts QLIKE by 8.7-11.2% over own-history at every horizon ($p < 0.001$), a gain seven times any graph term. We situate the negative result against eight recent methods reimplemented under the same protocol, none of which beats the baseline on QLIKE, and one of which owes its published gain to test-set early stopping in its official code. The eighth method is a faithful GNNHAR, a learned multi-hop message-passing graph network: its learned message passing still does not beat own-history on QLIKE on either market ($p = 0.000$). The graph null replicates on a second market, 405 Vietnam HOSE tickers, where the earnings lever does not transfer, which localizes the earnings gain to the S&P 500 while the graph null holds on both markets.

Keywords: Volatility forecasting · HAR · Graph neural network · Earnings · Diebold–Mariano.

1 Introduction

Daily volatility prices options and margin. The Heterogeneous Autoregressive (HAR) model [1] forecasts realized volatility from three own-history lags and remains the benchmark. A recent literature argues that HAR treats each stock in isolation and misses cross-firm spillover, and adds graph neural networks that aggregate a stock’s correlated or spillover-linked neighbors. Reported gains reach 8-50% on squared-error metrics.

We test whether cross-firm structure helps once the comparison is fair. A market factor already carries the common component of cross-stock co-movement, so the question is not “does a graph beat a stock in isolation” but “does a graph add anything beyond own-history and a market factor.” We express the market factor itself as a graph: the uniform, fully-connected adjacency whose neighbor aggregate is the cross-sectional mean. A correlation graph, a sector graph, and a random-edge placebo then differ from it only in their adjacency, which isolates the value of the wiring.

Under this framing, on the S&P 500, the correlation graph adds nothing a market factor does not already provide. It ties the uniform market graph at every horizon and beats a random-edge placebo at none. A small short-horizon graph signal survives only when the model’s own-history is stripped to three HAR lags; adding the stock’s own log-volatility momentum features erases it. The cross-sectional neighbor signal is redundant with the stock’s own past.

The lever that beats HAR is forward-looking and specific to the firm. A scheduled earnings release is known weeks ahead and raises volatility around the announcement. A QLIKE-loss gradient-boosting model that encodes the distance from the forecast target to the nearest scheduled release cuts QLIKE by 8.7-11.2% over own-history at every horizon, seven times the size of any graph term, and a decomposition assigns the bulk of the gain to earnings rather than to nonlinearity or added features.

This paper contributes three results. First, under a leakage-controlled, placebo-controlled, market-factor-as-graph protocol, cross-firm graph structure adds no reliable incremental QLIKE value for daily S&P 500 volatility at any horizon, and the apparent short-horizon signal is redundant with own-history momentum. Second, forward-looking scheduled earnings are the dominant lever, and we decompose the gain. Third, eight recent graph and cross-market methods reimplemented under the same protocol fail to beat the baseline on QLIKE, and one method’s published gain is a test-set early-stopping artifact. The eighth method is a faithful GNNHAR whose learned multi-hop message passing still does not beat own-history on QLIKE on either market. A replication on 405 Vietnam HOSE tickers confirms the graph null on a second market and shows the earnings lever is specific to the S&P 500 and does not transfer to Vietnam under real announcement dates.

2 Setup and Baselines

Target and metric. We forecast daily Parkinson variance, $\text{pk}_t = \ln(H_t/L_t)^2/(4 \ln 2)$ [2], at horizons $h \in \{1, 5, 10, 22\}$ over 497 S&P 500 constituents. We report QLIKE [3] as the primary loss, with MSE, RMSE, MAE, and R^2 . We test significance with a Diebold–Mariano statistic [4] on per-trading-date mean loss, because within-day stock errors are dependent.

Baselines. HAR fits three own-history lags by ordinary least squares. HARQ adds a realized-quarticity interaction. The own-history gradient-boosting model (GBM) trains a gamma-deviance regressor, equal to QLIKE up to a constant, on

Table 1. QLIKE, S&P 500 graph variants. No model is DM-distinguishable from the market aggregate, so no value is bolded. Bold marks the QLIKE that is significantly best by a date-clustered Diebold–Mariano test; unbolded values are not DM-distinguishable from the best.

h	GBM (own)	+market	+corr	+placebo	corr vs market	corr vs placebo
1	0.3570	0.3539	0.3536	0.3546	+0.07% ($p=.84$)	+0.26% ($p=.13$)
5	0.4116	0.4127	0.4108	0.4116	+0.46% ($p=.47$)	+0.20% ($p=.60$)
10	0.4307	0.4344	0.4328	0.4306	+0.37% ($p=.53$)	−0.50% ($p=.15$)
22	0.4464	0.4605	0.4495	0.4512	+2.39% ($p=.05$)	+0.37% ($p=.56$)

nine own-history features: three HAR lags, a realized-quarticity proxy, and five log-volatility momentum terms. No market or volume scalar enters the feature set; cross-sectional information enters only through graph aggregates.

Market factor as a graph. A neighbor aggregate under a uniform adjacency, $g_{\text{mkt}}[i, t] = \frac{1}{N-1} \sum_{j \neq i} \text{pk}[j, t]$, equals the cross-sectional mean and serves as the market factor. A correlation graph keeps each node’s top-10 train-correlation neighbors with positive row-normalized weights; a sector graph uses same-sector membership; a placebo rewires 10 random neighbors per node, averaged over three seeds. Each graph contributes one neighbor-aggregate feature, so the models differ only in the adjacency.

Protocol. We use expanding-window walk-forward folds with a target-horizon embargo. All correlations, adjacencies, factor estimates, and model fits use training data only; the test window is read once. Every gradient-boosting model averages three seeds. Every graph is compared against the random-edge placebo of matched degree.

3 Cross-Firm Graphs Add No Incremental Value Beyond Own-History and a Market Factor

Table 1 reports QLIKE for the own-history GBM and its graph-augmented variants on the S&P 500.

The correlation graph ties the uniform market graph at every horizon and beats the random-edge placebo at none. Its only edge over the market graph, at $h22$, coincides with the market graph itself hurting, so the correlation graph recovers ground the market aggregate loses rather than adding signal.

A sector graph and an explicit conditional test reach the same null. Table 2 adds an industry-sector adjacency, whose neighbor aggregate averages same-sector firms, and the direct Own+Market+Corr comparison.

The sector graph ties the uniform market graph at both short horizons ($h1$ +0.22%, $p = 0.43$; $h5$ +0.36%, $p = 0.55$). Adding the correlation graph on top of the market factor moves QLIKE by at most 0.20% ($h1$ $p = 0.10$; $h5$ $p = 0.41$), so the correlation graph carries no incremental value the market factor does not already hold, and an economically motivated sector wiring does no better.

Table 2. QLIKE, S&P 500 sector and conditional test. No model is DM-distinguishable from the market aggregate, so no value is bolded. Bold marks the QLIKE that is significantly best by a date-clustered Diebold–Mariano test; unbolded values are not DM-distinguishable from the best.

h	GBM (own)	+market	+sector	+corr	+market+corr
1	0.3570	0.3539	0.3531	0.3536	0.3532
5	0.4116	0.4127	0.4113	0.4108	0.4121

The null holds on every metric, not QLIKE alone. Table 3 reports QLIKE, RMSE, MAE, and R^2 for all seven models at each horizon. The correlation graph never significantly beats the market aggregate on any metric (DM QLIKE $p = 0.84/0.47/0.53/0.05$ at $h1/h5/h10/h22$), and adding the market aggregate even raises RMSE and lowers R^2 at $h5$ and $h10$, thin evidence that the aggregate adds noise at the longer horizons rather than signal. RMSE and MAE are scaled by 10^{-4} .

The short-horizon signal is redundant with own-history momentum. On a minimal three-lag HAR base, the correlation graph beats the market graph at $h1$ by 0.80% ($p = 0.035$), because the bare HAR base lacks any cross-sectional or momentum information. Restoring the five log-volatility momentum features to the base collapses the same comparison to +0.07% ($p = 0.84$). The stock’s own momentum features carry what the neighbor aggregate would otherwise supply.

An oracle bound locates the missing signal in time. Replacing the neighbor value at the origin t with the neighbor value at the target day $t + h$, an illegal upper bound contemporaneous with the label, cuts QLIKE by 19.2-27.2% over the correlation graph ($p < 0.001$ at every horizon) where the legal graph cuts it by under 1%. The cross-sectional signal exists and is large, but it arrives contemporaneously with the target, not at the forecast origin, so no origin-time graph can capture it.

Takeaway. Conditional on own-history and a market factor, cross-firm graph structure adds no reliable incremental QLIKE value at any horizon on the S&P 500. The signal a graph could carry is either redundant with the stock’s own momentum or contemporaneous with the target.

Table 3. All metrics, S&P 500. RMSE and MAE in units of 10^{-4} . This is a graph-null comparison, so no value is bolded. Bold marks the QLIKE that is significantly best by a date-clustered Diebold–Mariano test; unbolded values are not DM-distinguishable from the best.

h	Model	QLIKE	RMSE	MAE	R^2
1	HAR	0.3697	7.586	2.132	0.158
1	HARQ	0.3780	7.515	2.151	0.174
1	GBM	0.3570	7.454	2.089	0.187
1	+market	0.3539	7.423	2.075	0.194
1	+corr	0.3536	7.460	2.083	0.186
1	+sector	0.3531	7.435	2.077	0.191
1	+placebo	0.3546	7.437	2.076	0.191
5	HAR	0.4364	7.836	2.297	0.100
5	HARQ	0.4377	7.831	2.309	0.101
5	GBM	0.4116	7.821	2.255	0.103
5	+market	0.4127	7.977	2.302	0.068
5	+corr	0.4108	7.882	2.279	0.090
5	+sector	0.4113	7.887	2.281	0.088
5	+placebo	0.4118	7.924	2.284	0.080
10	HAR	0.4615	7.883	2.356	0.093
10	HARQ	0.4614	7.877	2.358	0.094
10	GBM	0.4307	7.913	2.312	0.085
10	+market	0.4344	7.959	2.316	0.075
10	+corr	0.4328	7.984	2.344	0.069
10	+sector	0.4316	7.973	2.337	0.072
10	+placebo	0.4310	8.002	2.337	0.065
22	HAR	0.4855	7.954	2.419	0.079
22	HARQ	0.4831	7.944	2.415	0.081
22	GBM	0.4464	7.895	2.359	0.092
22	+market	0.4605	7.877	2.369	0.096
22	+corr	0.4495	7.894	2.383	0.093
22	+sector	0.4496	7.869	2.377	0.098
22	+placebo	0.4489	7.860	2.376	0.100

4 Forward-Looking Earnings Are the Lever

A diagnostic locates the missing lever. The own-history GBM beats HAR-X across the calm deciles, yet its remaining QLIKE concentrates in the storm decile: decile D9 alone carries 43% of the model’s total test QLIKE (panel B). The model under-forecasts exactly there, where its median forecast-to-realized ratio falls below one while every calm decile sits above one (panel C). Permutation importance ranks the three own-history HAR lags far above every momentum and auxiliary term (panel E), so own-history is exhausted and the storm component needs a signal from outside the stock’s past. A scheduled earnings release is known weeks ahead and drives the announcement-driven part of these spikes, which makes a forward-looking earnings feature the natural lever for the residual the GBM cannot reach.

Table 4 adds a scheduled-earnings feature to the own-history GBM.

Table 5 reports all metrics for the own-history GBM, the earnings model, and the earnings-plus-graph model. The earnings feature improves every metric, not QLIKE alone. It posts the lowest RMSE and MAE and the highest R^2 at every horizon, lifting R^2 at $h1$ from 0.187 to 0.211, and the QLIKE gain over own-history is significant at every horizon (DM $p = 0.000$). RMSE and MAE are scaled by 10^{-4} .

The earnings feature cuts QLIKE by 8.7-11.2% over own-history at every horizon, seven times the largest graph term and significant at every horizon. Against the HAR baseline the earnings model gains 14-16%. Nonlinearity and the QLIKE-loss alignment, isolated by the own-history GBM over HAR, contribute a further 3.4%. A graph added on top of earnings changes nothing at short horizons and hurts at the monthly horizon, so the two signals do not combine.

Earnings do not fix the storm tail. Decomposed by realized-volatility decile, the model’s error concentrates in the top decile, which carries roughly ten times the calm-day QLIKE, and its forecast-to-realized ratio there is about 0.4. The model compresses its predictions toward the conditional mean because the storm magnitude is absent from origin-time features. The residual is exogenous, which is why no cross-firm graph recovers it.

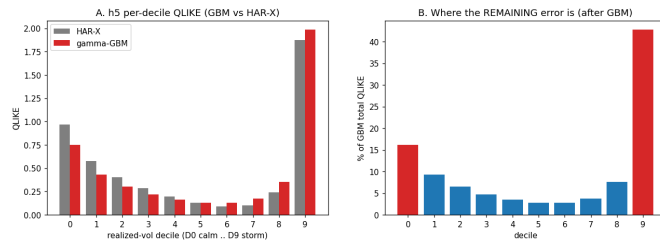


Fig. 1. Panel B: the own-history GBM beats HAR-X on the calm deciles, and its remaining QLIKE concentrates in the storm decile D9, which carries 43% of the model total.

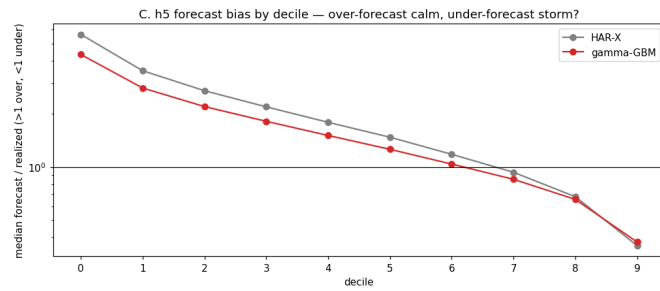


Fig. 2. Panel C: the GBM under-forecasts the storm deciles, where its median forecast-to-realized ratio drops below one.

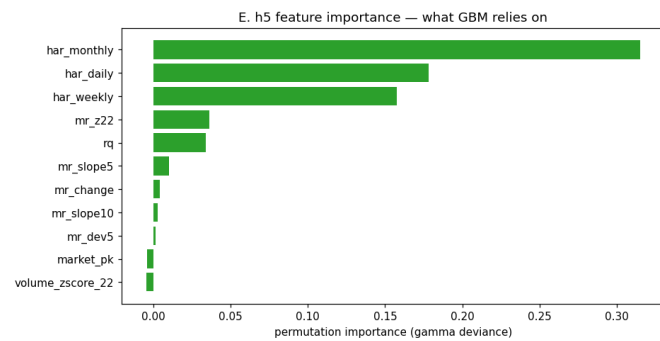


Fig. 3. Panel E: permutation importance ranks the own-history HAR lags far above every other feature, so own-history is exhausted.

Table 4. QLIKE, S&P 500 earnings. GBM+earn beats GBM at every horizon (DM $p < 0.001$). Bold marks the QLIKE that is significantly best by a date-clustered Diebold–Mariano test; unbolded values are not DM-distinguishable from the best.

h	GBM (own)	GBM+earn	earn vs GBM	+earn+graph vs +earn
1	0.3570	0.3172	+11.2% ($p < .001$)	+0.42% ($p=.20$)
5	0.4116	0.3705	+10.0% ($p < .001$)	−0.26% ($p=.46$)
10	0.4307	0.3912	+9.2% ($p < .001$)	−0.68% ($p=.06$)
22	0.4464	0.4074	+8.7% ($p < .001$)	−0.90% ($p < .001$)

Table 5. Earnings across all metrics, S&P 500. RMSE and MAE in units of 10^{-4} . Bold marks the QLIKE that is significantly best by a date-clustered Diebold–Mariano test; unbolded values are not DM-distinguishable from the best.

h	Model	QLIKE	RMSE	MAE	R^2
1	GBM	0.3570	7.454	2.089	0.187
1	GBM+earn	0.3172	7.346	2.017	0.211
1	GBM+earn+corr	0.3158	7.365	2.010	0.207
5	GBM	0.4116	7.821	2.255	0.103
5	GBM+earn	0.3705	7.728	2.180	0.125
5	GBM+earn+corr	0.3715	7.787	2.196	0.111
10	GBM	0.4307	7.913	2.312	0.085
10	GBM+earn	0.3912	7.822	2.239	0.107
10	GBM+earn+corr	0.3939	7.899	2.272	0.089
22	GBM	0.4464	7.895	2.359	0.092
22	GBM+earn	0.4074	7.809	2.277	0.112
22	GBM+earn+corr	0.4110	7.802	2.300	0.114

5 Recent Graph and Cross-Market Methods Do Not Survive

We reimplement eight recent methods under a common daily-OHLC information regime and score each against the baseline: H-ETE-GNN (transfer-entropy country-ETF graph), DCRNN-HAR (dynamic Diebold–Yilmaz spillover [5]), ASTGCN (attention spillover convolution), MSub-GNN (trading-news-candlestick fusion), an influence-weighted-news model, a GNN survey’s feasible edge constructions, a technical-indicator battery, and GNNHAR (a learned multi-hop message-passing graph network). None beats the baseline on QLIKE with DM significance. Where a method wins on squared error it ties or loses on QLIKE, and it does not beat its random-edge placebo. MSub-GNN’s fusion never beats its trading-only branch; the influence-weighted-news model never beats equal weighting, the paper’s own claim; technical indicators add at most 0.63%.

DCRNN-HAR reports that it surpasses HAR. Its official code selects the training epoch by the test-set loss. Under a chronological validation split, HAR beats DCRNN-HAR on QLIKE at $h1$ and $h5$ ($p < 0.001$) and ties at $h22$, and the dynamic graph never beats a static graph or a no-graph control. The published gain is a test-set early-stopping artifact.

GNNHAR: a learned multi-hop GNN does not exploit cross-firm structure beyond own-history plus a market factor. The eighth method is a faithful re-implementation of GNNHAR [7] (Zhang, Pu, Cucuringu and Dong, IJF 2024, arXiv:2308.01419; official repo [chaozhang-ox/GNNHAR](https://github.com/chaozhang-ox/GNNHAR)), a learned multi-hop message-passing GNN carrying the paper’s own baselines HAR and GHAR (linear graph-HAR). It answers the reviewer objection directly: a real GNN, not a scalar market factor, gets the chance to exploit cross-firm structure. We run it under our protocol, walk-forward with QLIKE and date-clustered DM, a three-seed ensemble over eight folds, on both markets, at all horizons, across variants that cross $\{1L, 2L\}$ depth with $\{\text{correlation graph, no-graph } A = I \text{ control, random-edge placebo}\}$ adjacency and $\{\text{HAR3, OWN9}\}$ node features. On the S&P 500 the GNN never beats the own-history GBM: the GBM posts QLIKE 0.357/0.412/0.431/0.447 at $h1/h5/h10/h22$ while the matched GNNHAR2L-corr-OWN9 posts 0.368/0.434/0.458/0.471, and DM GNNHAR2L-corr-OWN9 vs GBM runs $-2.96\% / -5.49\% / -6.41\% / -5.34\%$ with $p = 0.000$ at every horizon in the GBM’s favor; the GNN also loses to GBM+corr ($p = 0.000$). The correlation graph does not help inside the GNN and hurts at longer horizons: GNNHAR2L-corr-HAR3 vs the no-graph control GNNHAR2L-none-HAR3 runs $-0.32\% / -3.96\% / -5.40\% / -3.53\%$ ($p = 0.60/0.000/0.000/0.000$), a tie at $h1$ and the no-graph control significantly better at $h5/h10/h22$. GHAR, the paper’s linear graph-HAR baseline, is worse than HAR (DM GHAR vs HAR $-36.6\% / -27.6\% / -17.9\% / -16.3\%$), so a linear neighbour-augmented HAR does not fit the daily Parkinson target. Several correlation-graph GNN variants show per-seed QLIKE blow-ups (GNNHAR2L-corr-HAR3 per-seed 4.09 ± 17.8 at $h1$) that inflate the ensemble mean; the stable no-graph variant is the fair comparison and it still loses to the own-history GBM, because QLIKE’s sensitivity to under-prediction makes the learned GNN unstable under this protocol.

On HOSE the GNN loses to the own-history GBM and turns markedly unstable on the thin market: the GBM posts QLIKE 1.569 ($h1$) and 1.649 ($h5$), the no-graph GNNHAR2L-none-HAR3 posts 1.579/1.656 (it loses to the GBM), and correlation-graph variants blow up (GNNHAR2L-corr-OWN9 ensemble 40.17 at $h1$ with `fit_diagnostics=overfit` and individual test folds reaching QLIKE 179-432); DM GNNHAR2L-corr-OWN9 vs GBM runs -1.43% at $h5$ ($p = 0.000$). The GNN beats HAR on HOSE (DM $+7.98\%$ at $h1$) but never the own-history GBM, and the $h10$ and $h22$ folds carry the same ordering. On HOSE the learned graph GNN adds instability, not signal, consistent with the thin-market spike behaviour reported in Section 7. GNNHAR confirms the paper’s central negative result: a learned multi-hop GNN does not exploit cross-firm structure beyond own-history plus a market factor on either market. The full per-horizon GNNHAR result JSONs, carrying all five metrics and train/val/test fit evidence, are produced by the Colab A100 notebook and committed under `results/gnnhar/`.

6 Related Work

Graph volatility forecasting. GNNHAR [7], STG-Spillover, and the methods of Section 5 wire stocks or markets through correlation or spillover edges and report squared-error gains. GNNHAR itself finds that multi-hop neighbor information gives no clear advantage and credits its gains to nonlinearity and the QL loss. A regime-dependent temporal graph attention network reports gains on squared and absolute error using Diebold–Yilmaz spillover edges, the same edges a proportional loss and a no-graph control do not reward here. Prior Vietnamese work builds correlation and centrality networks to predict index volatility at a monthly horizon without a HAR or QLIKE benchmark. We extend this literature to a placebo-controlled, market-factor-as-graph test on daily per-firm QLIKE and show the gains do not survive.

HAR and its extensions. HAR [1], VHAR, and HARQ [6] model volatility from own history and linear spillover. We keep HAR and HARQ as baselines and show a QLIKE-loss gradient-boosting model with earnings dominates them without a graph.

Earnings and volatility. Studies document that earnings announcements raise implied and realized volatility. We operationalize the effect as a forward-looking forecasting feature and quantify its contribution against the strongest own-history and graph baselines.

7 Cross-Firm Graphs and Earnings on the Vietnam Market (HOSE)

The graph null generalizes to a second market and the S&P 500 earnings lever does not. On 405 Ho Chi Minh Stock Exchange (HOSE) tickers under the identical protocol, a correlation graph again ties the market factor, and the forward-looking earnings feature that cuts S&P 500 QLIKE 9-11% produces no incremental value on HOSE.

Setup. We rebuild the panel on 405 HOSE tickers, forecasting daily Parkinson variance at the same four horizons under the same walk-forward, QLIKE-scored, date-clustered Diebold–Mariano protocol, with 390,577 to 399,040 stock-day observations per horizon.

The own-history GBM wins and the graph adds nothing. GBM posts the lowest QLIKE and the highest R^2 at every horizon (QLIKE 1.5679/1.6478/1.6842/1.7263, R^2 0.198/0.133/0.107/0.073), while HAR and HARQ trail by a wide margin (QLIKE 1.79 to 1.84, R^2 near zero to 0.05). Table 6 reports all metrics for the eight models. The correlation graph ties the market factor at every horizon (corr vs market DM p = 0.138/0.055/0.465/0.533), and the sector and placebo graphs land on the GBM QLIKE. The $h1$ GBM+market entry, QLIKE 2.5126, is a thin-market artifact rather than a result: one walk-forward fold spikes to 8.87 in the per-fold QLIKE where the cross-sectional mean turns unstable, and the remaining folds sit at the GBM level.

Earnings does not transfer to Vietnam. Using real crawled Vietnam announcement dates, GBM+earn ties or slightly worsens GBM at every horizon: QLIKE 1.5730 vs 1.5679 at $h1$ (earnings worse, DM p = 0.000), 1.6485 vs 1.6478 at $h5$ (p = 0.214), 1.6862 vs 1.6842 at $h10$ (p = 0.122), and 1.7262 vs 1.7263 at $h22$ (p = 0.919). The per-fold QLIKE shows no gain even in the 2025-2026 folds where Vietnam earnings coverage is comprehensive. The forward-looking earnings lever that cuts S&P 500 QLIKE 9-11% does not carry to HOSE. A per-stock, per-day diagnostic on the HOSE test set locates two compounding causes. First, the feature fires on only 5.6% of test observations, and 0.84% before 2025, so it cannot move the pooled metric across most of the panel. Second, where it fires the volatility response is muted: realized variance on announcement-proximate days runs 1.1 times the baseline, against the pronounced S&P 500 earnings spike, because Vietnam’s daily price limits cap the informational jump. Where the feature moves the prediction it worsens QLIKE on about 68% of those observations, adding variance with no volatility response to exploit. The damage concentrates in low-volatility utilities and in the 2025-2026 dense-coverage window.

The oracle bound confirms the signal is contemporaneous. Replacing the neighbor value at the origin with its value at the target day cuts QLIKE by 1.3%/2.6%/3.2%/2.4% over the correlation graph (DM p = 0.009/0.000/0.000/0.012). The gain is smaller than the S&P 500’s 19-27% and points the same way: the cross-sectional signal is contemporaneous with the label, not available at the forecast origin, so no origin-time graph captures it.

Earnings coverage. The Vietnam announcement dates combine the State Securities Commission (SSC) disclosure portal, comprehensive for 2025-01 through 2026-09, with the vnstock/VCI news feed, which spans 2016-2026 but stays sparse at roughly 218 tickers and is strongest on VN30 and VN100 names. The combined archive holds 5,723 dates across 402 tickers, dense for 2025-2026 and thin before. The no-transfer result holds in the well-covered 2025-2026 folds, so it does not rest on missing dates in that window; the pre-2025 earnings signal remains coverage-limited.

Takeaway. The graph null replicates on a second, structurally different market. The earnings lever is specific to the S&P 500 and does not generalize to HOSE under the available real announcement dates.

Table 6. All metrics, HOSE. RMSE and MAE in units of 10^{-4} . The own-history GBM is the best legal model at every horizon; the $h1$ GBM+market 2.5126 is a thin-market spike artifact, not the best. Bold marks the QLIKE that is significantly best by a date-clustered Diebold–Mariano test; unbolded values are not DM-distinguishable from the best.

h	Model	QLIKE	RMSE	MAE	R^2
1	HAR	1.8117	8.126	5.193	0.020
1	HARQ	1.7854	7.993	5.062	0.051
1	GBM	1.5679	7.348	3.944	0.198
1	GBM+earn	1.5730	7.401	3.966	0.187
1	GBM+market	2.5126	7.397	3.926	0.188
1	GBM+corr	1.5692	7.375	3.941	0.192
1	GBM+sector	1.5791	7.414	3.992	0.184
1	GBM+placebo	1.5739	7.434	3.947	0.180
5	HAR	1.8218	8.140	5.227	0.009
5	HARQ	1.8075	8.074	5.152	0.025
5	GBM	1.6478	7.613	4.265	0.133
5	GBM+earn	1.6485	7.613	4.276	0.133
5	GBM+market	1.6990	7.615	4.223	0.133
5	GBM+corr	1.6474	7.612	4.247	0.133
5	GBM+sector	1.6530	7.618	4.242	0.132
5	GBM+placebo	1.6527	7.632	4.252	0.129
10	HAR	1.8277	8.161	5.250	0.004
10	HARQ	1.8180	8.118	5.198	0.014
10	GBM	1.6842	7.724	4.425	0.107
10	GBM+earn	1.6862	7.721	4.441	0.108
10	GBM+market	1.6854	7.723	4.369	0.108
10	GBM+corr	1.6837	7.719	4.396	0.109
10	GBM+sector	1.6868	7.733	4.413	0.105
10	GBM+placebo	1.6851	7.731	4.418	0.106
22	HAR	1.8355	8.195	5.282	-0.002
22	HARQ	1.8299	8.171	5.250	0.004
22	GBM	1.7263	7.881	4.615	0.073
22	GBM+earn	1.7262	7.881	4.627	0.073
22	GBM+market	1.7298	7.879	4.546	0.074
22	GBM+corr	1.7276	7.882	4.591	0.073
22	GBM+sector	1.7272	7.880	4.598	0.074
22	GBM+placebo	1.7268	7.889	4.610	0.071

8 Limitations

The earnings feature measures distance to the nearest scheduled release from an announcement-date archive that records realized dates, not a point-in-time scheduling snapshot. A date-jitter check shows the gain does not depend on exact-date timing. Perturbing every earnings date by a uniform random offset and refitting, the QLIKE gain over own-history falls from 11.1% at $h1$ with true dates to 9.0% at plus or minus one trading day, 6.6% at plus or minus three days, and 4.5% at plus or minus five days, and stays positive at every horizon under every jitter. A release known only to within a business week still cuts QLIKE by about 4%, so the lever survives coarse timing. A point-in-time scheduling calendar remains to be added to control rescheduling within the forecast window. The eight-method reimplementation degrades some methods from intraday realized variance or paid implied-volatility inputs to daily Parkinson variance, so a null under degraded inputs is weaker than a null on the original inputs; we state each deviation. The graph analysis is conditional on the Parkinson daily-OHLC regime and does not speak to intraday or option-implied signals. The Vietnam earnings test draws on real announcement dates that are comprehensive only for 2025-01 through 2026-09, from the SSC disclosure portal, plus a sparse 2016-2026 vnstock/VCI news feed covering roughly 218 tickers. The no-transfer finding holds in the well-covered 2025-2026 folds, but the pre-2025 Vietnam earnings signal is coverage-limited, so we claim no incremental earnings value on HOSE under the available real dates rather than a definitive full-history null.

Regime and shock-detection methods do not transfer to this evaluation. A regime-switching HAR reports MSE gains of 8-12% around the COVID period, and change-point detectors flag structural breaks, but our walk-forward test begins in July 2022, so the COVID regime sits entirely in the training window and cannot be exploited out-of-sample. The reported gains are on squared error rather than QLIKE, and detection lags the break, which does not help forecast the onset of the exogenous shocks that fall in the test window, the 2022 bear market and the April 2025 tariff shock. A limit-lock hurdle model on VN30 and VN100 did not beat the baseline; winsorized or floored QLIKE, not shock detection, is the effective handling of the zero-range limit-lock days that dominate the raw loss.

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